

Data Handling & Regression – Assignment 2 – Mountains vs beaches

This assignment uses the dataset as given in `mountains_vs_beaches_extended.csv`. This dataset is different from the dataset `mountains_vs_beaches.csv` you can find online.

Perform the exercises given below in either MATLAB or Python. Upload in Portflow your code (in the formats `.m`, `.mlx`, `.py`, `.ipynb`) and a concise report stating your methods, results and conclusions (in `.pdf`-format, or in the `.mlx`- or `.ipynb`-file).

Exercises

1. Load and clean the data.
1. Investigate what the data looks like. For example, make a few scatter plots of one variable vs another, and comment on their shape.
2. Find combinations of variables which can be modelled in the following ways:
 - a. an explanatory variable and an observational variable which can be modelled properly with a linear model.
 - b. an explanatory variable and an observational variable which, if converted to aggregated data, can be modelled properly with a linear model.
 - c. an explanatory variable and an observational variable which can be modelled properly with a logistic regression model.
3. Perform the fits. Comment on the quality of the fits.
4. Test the significance of the fits.
5. Build a model for the observational variable `Preference` using a feature selection method. Explain your choices and how you got to the final model.
6. Using classification trees and/or classification forests, build an alternative model. Make the model easily explainable to others. Also for this model, quantify the quality.
7. Find out how your method of making classification trees / forests determines the quality of the trees and how it builds the tree models.